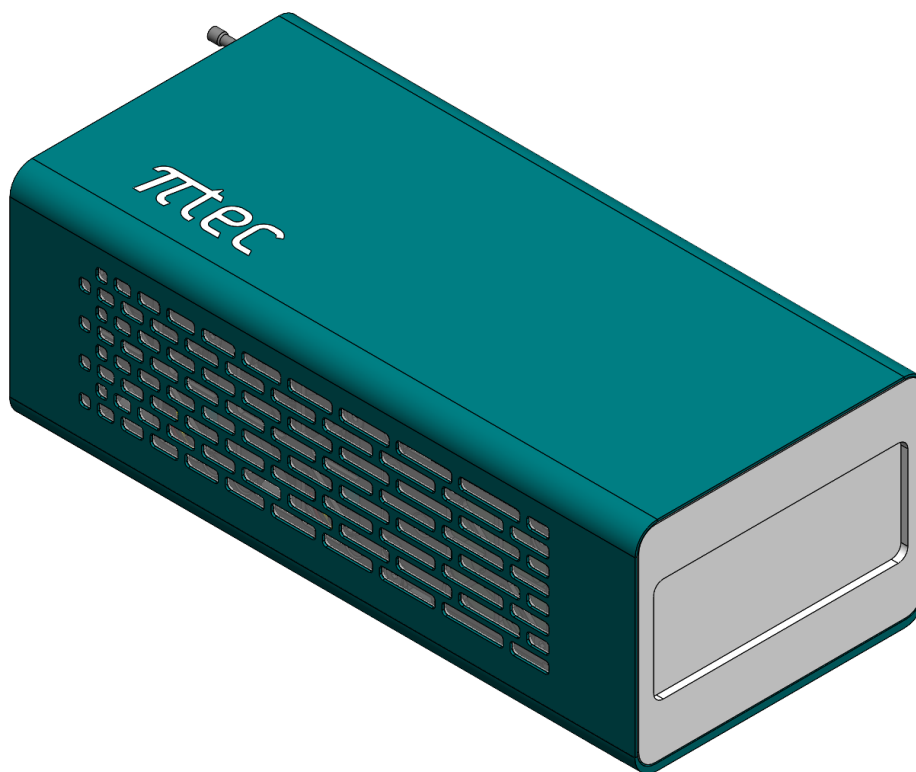


Compact X-ray Detector

RAD 800k - Si or CdTe

Installation User Manual



Revision History

Rev	Origination Date	Change Description	Originator(s)	Reviewed by
1.0	2024-01-03	Initial Release	Leticia Rosa Victor Araujo	
1.1	2024-14-06	Adjustments in installation procedure	Victor Araujo	Leticia Rosa
1.2	2024-09-08	Adjustments in installation procedure	Victor Araujo	
1.3	2024-11-27	Troubleshooting	Victor Araujo, Gabriel Amorim	
1.4	2025-10-27	Adjustments in installation procedure	Victor Araujo, Maria Medeiros	

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Safety Instructions

Before starting the system, make sure to read this manual and the technical specification to use and set up the detector safety. Failure to follow the instructions and warnings in this document may result in damage to the product.

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1. Introduction

RAD 800 k is a compact area detector for X-Ray applications based on the latest noise-free, photon counting hybrid-pixel technology (Figure 1). It is made with six Medipix3RX ASICs of $55 \mu\text{m}^2$ pixel size, resulting in **786 kPixels** of resolution, with an active area of **33.2 x 87.3 mm²**. This detector offers high resolution, high speed and high efficiency in a small form factor – suitable for several techniques and applications required for synchrotrons.

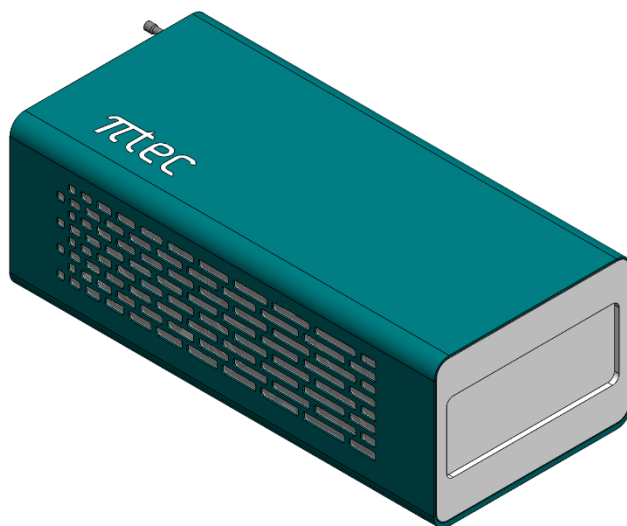


Figure 1: Compact Area Detector - RAD.

2. Transport Case Content

The detectors are shipped in a reinforced transport case, shown in Figure below. The detector is kept in place by high density foam on all sides. To open the top lid of the case, unlock all the clasps. All the necessary components for the detector operation are contained within the transport case along with the detector. A list of the transport case content is shown in the table below.

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Figure 2: Detector transport case closed (left) and opened (right).

Component		RAD 800 K
1	Compact Area Detector	
2	Head Protector	1 x
3	Mellanox ConnectX -5 NIC 10/25GbE Dual-Port SFP28 MCX512A-ACAT	2 x
4	SFP+ Optical Transceiver Module 10GBASE-SR 850nm 300m MFM1T02A-SR	4 x
5	LC UPC to LC UPC Duplex OM4 Multimode PVC (OFNR) optical cable	2 x
6	Power supply SDM120-12-U-P5	1 x

Table 1: RAD transport case content.

Note	
	A server for detector controlling may be included (if contracted) in an extra transport case.

3. Hardware Specifications

3.1. Sensor Specifications

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There are two detector models available, RAD 400K that has a resolution of 393 kpixels composed by one strip of Medipix3RX sensors, and RAD 800K that has a resolution of 786 kpixels with two strips of sensors. The active area of each detector model and the respective sensor specifications are shown in Table 2 and illustrated in the Figures below.

RAD 800K	
	
Resolution [kpixels]	786
Pixel Size [μm]	55
Active Area [mm^2]	33.2 x 87.3
ASIC Size [mm^2]	14.1 x 15.88
Sensor Array [chips]	2 x 6
Pixel Array	512 x 1536
Gap* [mm]	0.06
Gap** [mm]	0.45

Table 2: RAD sensor array characteristics.

*Vertical Gap between sensor rows.

** Horizontal Gap between ASIC chips for CdTe (single modules). Silicon detectors are made of a monolithic sensor block, and therefore the entire area is active.

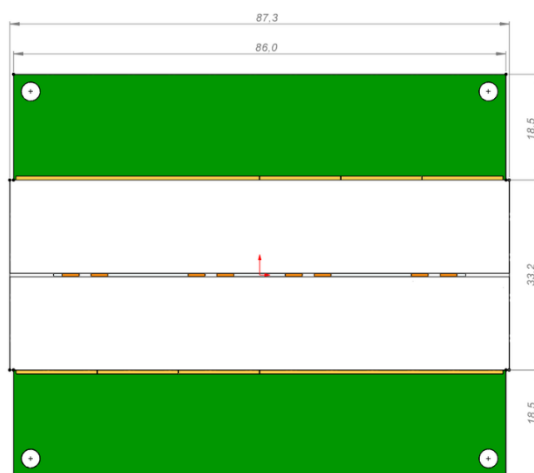


Figure 3: RAD 800K illustration with dimensions of active area for Silicon 300 μm thick edgeless sensor.

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The sensor array can have different characteristics depending on the type of semiconductor applied. Silicon (Si) arrays are made of a monolithic silicon block bonded over 6 Medipix chips. This structure is known as one Hexa. Cadmium Telluride (CdTe) arrays are made of six single blocks of CdTe bonded to six Medipix chips. The available sensor thicknesses for Si are 300 μm and 675 μm , and for CdTe 1000 μm . Medipix3RX chips themselves have a structure of 256×256 pixels which form a square matrix with a pitch of 55 μm . The sensor specifications for each semiconductor material are shown in Table 3.

Item	Silicon	CdTe
Sensor Type	P on N	P on N Ohmic
Sensor Array Design	1 Hexa	6 Single sensors
Standard Pixel Area [μm^2]	55	
Sensor Thickness [μm]	300 or 675	1000
Sensor Edge	Edgeless or Guard Ring	Edgeless
Depletion Voltage [V]	< 80 @ 300 μm < 120 @ 675 μm	< -300
Guard Ring Width	0.327 mm (not active)	-

Table 3: Sensor Specifications.

3.2. Installation Requirements

The installation requirements for the RAD Detector Modules are described in the following Table.

Item	Requirement	Observations
RAD 800K		
Detector Power Input	1 x 12 VDC 20A	Commercial power supplies can be used
External trigger Input	TTL 3.3 - 5V	1 trigger input. SMA connectors
External trigger Output	TTL 5V Capable to drive 50 Ω Load	1 trigger output. SMA connectors.

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Data Interface	10/100/1000 ethernet interface based QSFP28-SR4-100G	MTP/MPO OM4+ 12 (six pairs) CROSS multimode FIBERS (MM) 50/125um optimized laser, for 100Gbps. Example of cable: MTF - 12F-30-LSZH<50/125 OM4>
Data Interface Length	100 m (maximum)	

Table 4: Detector installation requirements.

3.3. Detector Back Pannel Connectors

The back panel of RAD detectors are shown in Figure 4. The following table describes each connection.

Figure 4: Detector back panel connections RAD 800K.

Item	Requirement
RAD 400K & 800K	
TRIGGER IN	Trigger input
TRIGGER OUT	Trigger output
USB C	USB C communication
POWER IN 12v	6 pins power connector
ON/OFF BUTTON	Turn on/off button
10G SFP 1	Data output and control input through fiber optic 1
10G SFP 2	Data output and control input through fiber optic 2
BIAS IN	Bias supply
NITROGEN IN	Nitrogen Input for the module.

Table 5: Detector back panel hardware requirements.

3.3.1. LED System

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The detector is equipped with four different LEDs to show the status of power, connection, and environment conditions. The specifications are described in the table below. The respective LEDs position are indicated in Figure 5.

LED		Status		
		OFF	Green	Red
1	Power	Power supply is OFF	Power supply is ON	-
2	Ready	System NOT READY	System is READY	Initialization FAILED
3	Link Activity	Detector is not connected to the server	[Solid] - Detector and server are connected through ethernet link [Blinking] - Data transmission is active	FAILED in establishing connection and data transmission
4	Environment	-	Ambient temperature and humidity are within the operation conditions	Ambient temperature or humidity are out of operation conditions

Table 6: Detector LED System.

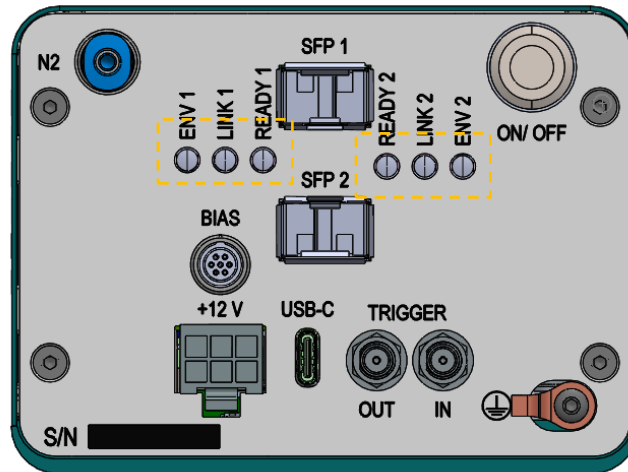


Figure 5: LED schematic position on detector's back panel.

3.3.2. Trigger Connections

RAD trigger connections are composed of one trigger input and one **processed output** available for synchronization with external hardware. There are a few modes available to use hardware triggering to define the active period of the detector. It is possible to define either single frames or multiple frames in an acquisition, or to launch a burst of frames after the receipt of a hardware trigger. The trigger modes are described as follows:

- **Trigger to Start a Frame:** This mode is set to run the system in a state where an external pulse triggers the start of Frame acquisition.
- **Trigger to Stop a Frame:** This will set the system to run with the shutter held open until a trigger is received. The trigger start will be executed by the internal trigger, and the Trigger Stop, rising or falling edge of the TTL signal, is activated after an external trigger signal is received.

3.3.3. Power Supply Connection

Although any commercial power supply within the specifications described may be used, it is recommended to use the one included with the detector.

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To power the detector, simply plug the power supply shown in Figure 8 to the detector's "POWER +12V" port illustrated in Figure 4. If connected correctly, when pressing the "ON/OFF" button the device power LED lights will turn on indicating the detector is on.



Figure 8: RAD Power supply.

4. Mechanical Specifications

4.1. Detector Dimensions

The detector dimensions and weight are shown in the table below. Figure 9 illustrates the design.

RAD 800K	
	
Width [mm]	105
Height [mm]	80
Length [mm]	235
Weight [kg]	2.1

Table 8: Detector dimensions.

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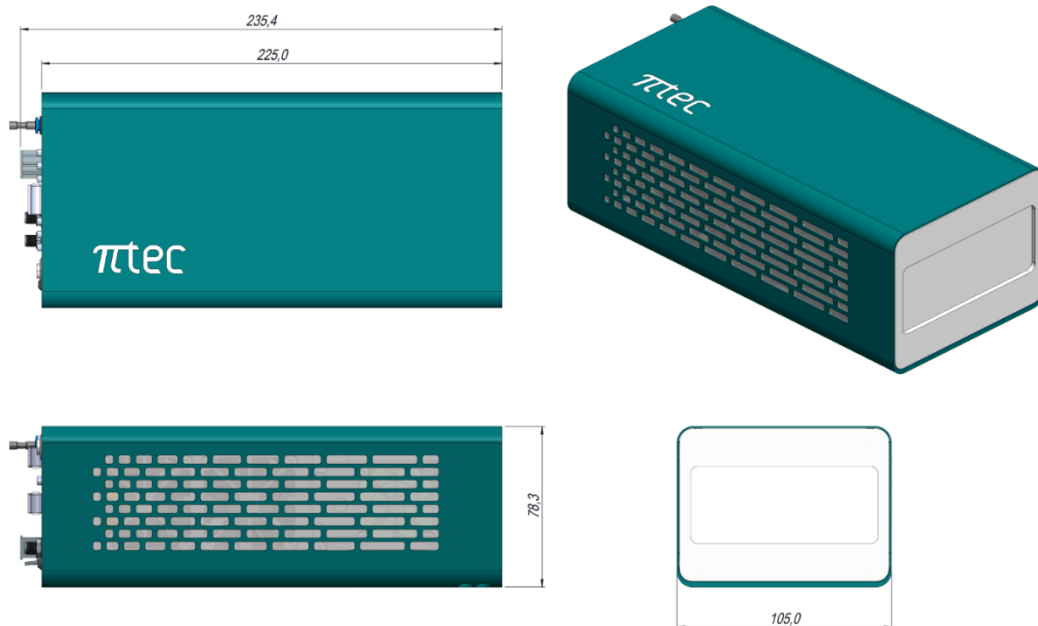
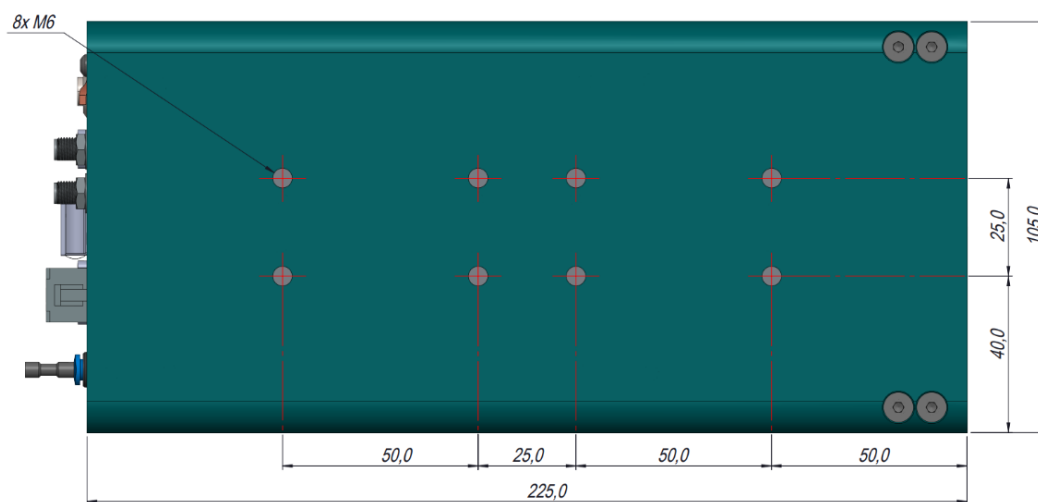


Figure 9: Illustration of RAD dimensions (up) RAD 800K.

4.2. Detector's Fasten Principle

The detector can be mounted from the bottom plate using the screw holes matrix to attach the detector to tables or supports for safe operation. The matrix is composed of 8 holes of **M6 x 10 mm** screws as shown in Figure 10. This Mounting piece is the base plate of the detector. Use at least 2 screws for each region: front, middle and back, for a safe attachment.



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Figure 10: Base plate of the detector with fastens holes.

4.3. Installation Requirements

4.3.1. Nitrogen Flow

To avoid sensor degradation due to moisture, the detector needs to be connected to a continuous flow of gaseous nitrogen when it is not in use (powered off). The Nitrogen flow requirements are specified in the table below. A schematic suggestion of installation using a 9 m³ cylinder is shown in Figure 11.

Item	Requirement
Nitrogen (N₂) pressure	0.05 bar (Maximum)
Pipes for external input of Nitrogen gas	Two ferrule fittings. External diameter = Ø6 mm

Table 9 Nitrogen supply requirements.

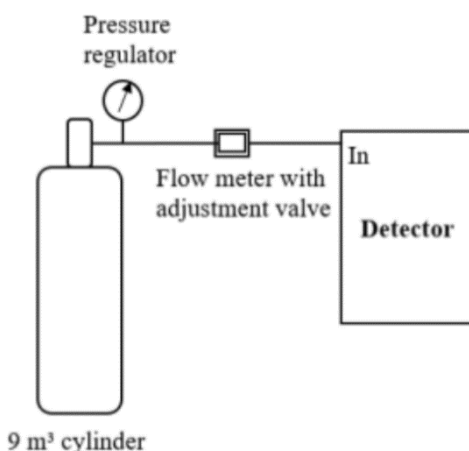


Figure 11: Suggested nitrogen setup using a 9 m³ nitrogen cylinder. Note: use a nitrogen filter.

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4.3.2. Cooling System

The detector is equipped with an air-cooling system. Therefore, no external cooling sources are required. A schematic view of the detector cooling region is shown in Figure 12.

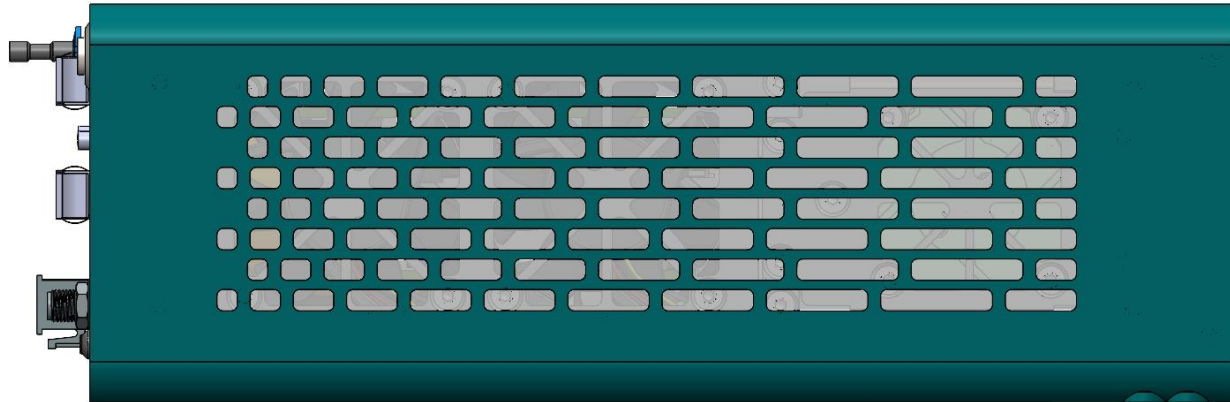


Figure 12: Detector lateral panel with cooling window for Air flow.

	<table><tr><th data-bbox="339 1096 1425 1098">Warning</th></tr><tr><td data-bbox="339 1098 1425 1104">Make sure not to obstruct the air flow by blocking the detector lateral walls where the heat is transferred from one side to the other.</td></tr></table>	Warning	Make sure not to obstruct the air flow by blocking the detector lateral walls where the heat is transferred from one side to the other.
Warning			
Make sure not to obstruct the air flow by blocking the detector lateral walls where the heat is transferred from one side to the other.			

5. Installation Quick Start Guide

This quick step by step installation manual will guide you through the process of the detector bring up, from the unboxing point until the fully operational stage.

5.1. Cable Connections

Connect all cables on the detector back panel plugs, following the order as indicated in next steps and illustrated in Figure 13.

1. Plug in the SFP 1 connector with the optical fiber cable on the SFP indicated plug.
2. Plug in the SFP 2 connector with the optical fiber cable on the SFP indicated plug.

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3. If applicable, connect the IN and OUT trigger connection cables on the indicated plugs.
4. Connect the inlet nitrogen flux hose on the indicated connector.
5. Connect the power supply cable on the +12 V connection.
6. Connect the USB-C cable.
7. Press and hold the power button until the green indicator light turns on.

After these steps, the detector is all set to be initialized.

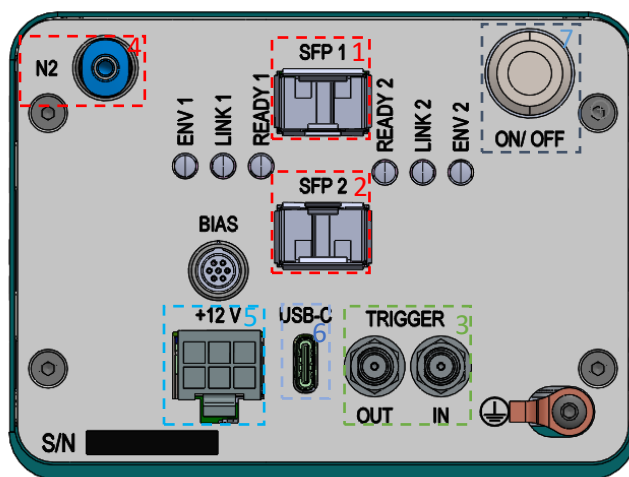


Figure 13: Cable connection order.

Warning	
	The bias connector (LEMO) must be connected, even when using the internal bias, to complete the circuit. Two LEMO connectors are included in the case.

5.2. Startup Procedure

Inspect the detector to make sure it is mounted correctly, and all cables are connected properly. Make sure the environment is within the specified conditions for the detector operation before proceeding.

8. Press the ON/OFF button to turn on the detector.
9. Wait at least 1 hour before start using the detector for thermal stabilization.

In this stage the power led indicates that the detector is on.

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**Warning**

Be careful not to turn on the detector when the temperature or humidity is outside of the specified conditions for operation.

5.3. System Initialization

5.3.1. Detector Initialization

Once the detector is up, it is necessary to establish its communication with the server. Make sure at this stage your server is properly configured and the detector control software is installed correctly as described in the Software Installation document provided. This is applicable if a server was not purchased with the detector. Otherwise, the server is provided with the software and configurations required.

The following steps are necessary to configure the detector connection for the first-time use. After is done, all configurations are saved, and it is not necessary to repeat the process in other occasions.

10. Open a shell terminal and consult the Ethernet port where the detector is connected which has “Up” status. Run the following command:

```
$ ibdev2netdev
```

See an output example below, where one detector connection is up and a second one is down. Make sure your connection is the two up to proceed.

```
pimega@severino:~$ ibdev2netdev
mlx5_0 port 1 ==> enp1s0f0 (Up)
mlx5_1 port 1 ==> enp1s0f1 (Down)
```

Figure 22: Terminal output for Ethernet connection check.

11. Consult the server IP address of the interface by running the command:

```
$ ip address
```

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Make sure the address number for the server is **10.255.255.1/30**. The figure below shows the configuration numbers for the server and the detector.

Server	Detector
10.255.255.1/30	-> 10.255.255.2

Figure 23: IP Configuration.

12. Check and make sure that the interface maximum transfer unit (MTU) is **4200**.
13. Make sure that the detector and server are in the same sub-network, which is mandatory.
14. Edit the configuration file for the Software Command Line Interface (CLI) communication interface. In the terminal, open the file as follows:

```
$ gedit ${~}/.config/pimegapss/config.json
```

Edit the file informing the detector Model, the Serial Number (optional), the Ethernet Port and the IPAddress. In case the S/N is unknown, consider a generic value like “RAD800k”. See the example below.

```
{
  "interface": {
    "Camera_Path": "",
    "Model": "RAD800k",
    "default_config_file": "/home/user/src/pimega-pss/init.templates/rad800k.ini",
    "modules": {
      "1": {
        "Ethernet_Port": "enp1s0f0np0",
        "IPs": "10.255.255.2",
        "Serial_Number": "mod1"
      },
      "2": {
        "Ethernet_Port": "enp1s0f1np1",
        "IPs": "10.255.255.6",
        "Serial Number": "mod2"
      }
    }
  }
}
```

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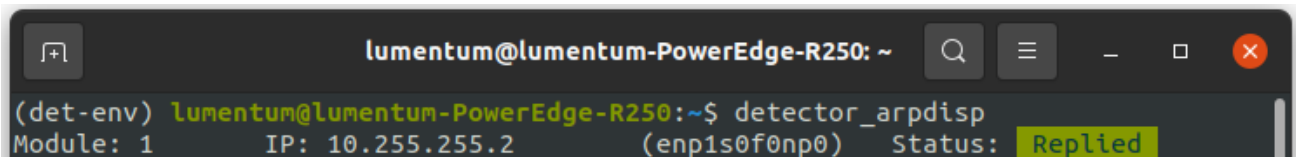
```
}  
}  
}
```

Figure 16: Configuration file.

15. Verify the detector-server communication status by running the command:

```
$ detector_arpdisp
```

You should get a reply as shown in the example image below, one line by module, in the case of the RAD 800k is two.



```
(det-env) lumentum@lumentum-PowerEdge-R250:~$ detector_arpdisp  
Module: 1      IP: 10.255.255.2      (enp1s0f0np0)  Status: Replied
```

Figure 24: Communication status reply.

When all modules are properly connected, the status will display **"Replied"** in green. If the message **"Failed"** appears in red, it indicates a connection issue with the detector. In such cases, please refer to the Troubleshooting section.

Now the "Link Activity" LED on the detector's backpanel should be green, indicating that the detector is all set and ready to be used.

6. Troubleshooting

6.1. IP Issues

In some cases, the server and detector may fail to connect properly. This is often caused by one of the modules having an incorrect IP address. The server is configured to identify both modules

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using different IPs, but this process can occasionally fail, requiring you to manually adjust the IP address of one of the modules.

Steps to Resolve IP Issues

1. Verify the Problem:

Open a terminal and run the following command

```
> ifconfig
```

A list of all connections will appear, as shown in the example image below.

The details include:

- The Mellanox port name for each module (highlighted in the orange rectangle).
- The server address (first group of numbers in red, indicated by the red rectangle).
- The subnet mask (in blue).
- The module's IP address associated with the connection port (in green).

```
ens3f0np0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 4200
    inet 10.255.255.1 netmask 255.255.255.252 broadcast 10.255.255.3
    inet6 fe80::63f:72ff:fedc:40cc prefixlen 64 scopeid 0x20<link>
    ether 04:3f:72:dc:40:cc txqueuelen 1000 (Ethernet)
    RX packets 12 bytes 2240 (2.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 241 bytes 24365 (24.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ens3f1np1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 4200
    inet 10.255.255.5 netmask 255.255.255.252 broadcast 10.255.255.7
    inet6 fe80::63f:72ff:fedc:40cd prefixlen 64 scopeid 0x20<link>
    ether 04:3f:72:dc:40:cd txqueuelen 1000 (Ethernet)
    RX packets 3047 bytes 268838 (268.8 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2323 bytes 258522 (258.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

For the RAD 800k detector, the server and detector must follow a specific pairing, such as Server 1 with Detector 3 or Server 5 with Detector 7, where each module is assigned to its own connection.

2. Correct Mismatched IPs:

If any IP address does not match the expected configuration:

- Connect the detector to the server using a USB-C cable.
- Open a terminal and run:

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```
> picocom -b115200 /dev/ttyUSB1
```

- This will open a shell interface, as shown in the image below. Each module is accessible via its USB number, typically **USB0** for the first module and **USB1** for the second. Press **Enter** to enable command input.

```
pimega@welbiserver:~$ picocom -b115200 /dev/ttyUSB0
picocom v3.2a

port is          : /dev/ttyUSB0
flowcontrol      : none
baudrate is      : 115200
parity is        : none
databits are     : 8
stopbits are     : 1
escape is        : C-a
local echo is    : no
noinit is        : no
noreset is       : no
hangup is        : no
nolock is        : no
send_cmd is      : sz -vv
receive_cmd is   : rz -vv -E
imap is          :
omap is          :
emap is          : crcrlf,delbs,
logfile is       : none
initstring       : none
exit_after is    : not set
exit is          : no

Type [C-a] [C-h] to see available commands
Terminal ready

shell>
```

3. If appear “??” on the shell interface

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```
(det-env) lumentum@dcu-server-03:~$ sudo picocom -b 115200 /dev/ttyUSB0
[sudo] password for lumentum:
picocom v3.1

port is      : /dev/ttyUSB0
flowcontrol  : none
baudrate is  : 115200
parity is    : none
databits are : 8
stopbits are : 1
escape is    : C-a
local echo is : no
noinit is    : no
noreset is   : no
hangup is    : no
nolock is    : no
send_cmd is  : SZ -vv
receive_cmd is : rz -vv -E
imap is      :
omap is      :
emap is      : crcrlf,delbs,
logfile is   : none
initstring   : none
exit_after is : not set
exit is      : no

Type [C-a] [C-h] to see available commands
Terminal ready
??
> ??
```

try the command go on the shell interface

```
> go
```

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```
(det-env) lumentum@dcu-server-03:~$ sudo picocom -b 115200 /dev/ttyUSB0
[sudo] password for lumentum:
picocom v3.1

port is      : /dev/ttyUSB0
flowcontrol  : none
baudrate is  : 115200
parity is    : none
databits are : 8
stopbits are : 1
escape is    : C-a
local echo is : no
noinit is    : no
noreset is   : no
hangup is    : no
nolock is    : no
send_cmd is  : SZ -vv
receive_cmd is : rz -vv -E
imap is      :
omap is      :
emap is      : crcrlf,delbs,
logfile is   : none
initstring   : none
exit_after is : not set
exit is      : no

Type [C-a] [C-h] to see available commands
Terminal ready
??
> ??
> go
```

After that the shell interface must accept new commands as show below

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```
TRACE manager/config/config_manager.cc:80 3.10.4: Adding config leaf: WarningTem
p
TRACE manager/config/config_manager.cc:80 3.10.5: Adding config leaf: CriticalTem
p
TRACE manager/config/config_manager.cc:80 3.10.0.0: Adding config leaf: tmp121_t
emp.0.0
TRACE manager/config/config_manager.cc:80 3.10.0.1: Adding config leaf: tmp121_t
emp.0.1
TRACE manager/config/config_manager.cc:80 3.10.0.2: Adding config leaf: tmp121_t
emp.0.2
TRACE manager/config/config_manager.cc:80 3.10.0.3: Adding config leaf: tmp121_t
emp.0.3
TRACE common/init.cc:79 Initialize the Control Manager
TRACE manager/config/config_manager.cc:80 0.0.0: Adding config leaf: control loo
p duration
TRACE manager/config/config_manager.cc:80 0.0.1: Adding config leaf: loop-too-sl
ow
Successful initialization
TRACE manager/control/contrTol_manager.cc:65 Starting control
TRACE manager/task/tasks.cc:82 Network task started
TRACE manager/task/tasks.cc:75 Shell task started

Shell (build: Mar 21 2025)
Copyright (c) 2024 Lumentum;
shell>
TFTP server start successful.
shell>
shell>
```

Now, try to verify the connection again using the *detector_arpdisp* command.

4. Verify the Connection:

In another terminal, run the command:

```
> detector_arpdisp
```

```
(det-env) lumentum@dcu-server-03:~$ detector_arpdisp
Module: 1      IP: 10.255.255.2      (enp23s0f0np0)  Status: Replied
Module: 2      IP: 10.255.255.6      (enp23s0f1np1)  Status: Replied
```

If both modules appear in **green**, the connection is successful, and you can proceed with the next commands as outlined in the instructions.

If the connection problem persists, use the description below.

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5. Check the Module's IP Address:

Use the following command to display the IP address of the selected module:

```
> ipconfig
```

```
shell> ipconfig
Eth#      : 0
Link      : off
MTU       : 1500
MAC       : 00:00:5e:13:00:1f
IP4       : 10.255.255.2 Type: DHCP
IP4 Mask  : 255.255.255.252
IP4 Gate  : 10.255.255.1
IP4 DNS   : 0.0.0.0
IP6       : DISABLED
Link status task stopped
shell>
```

6. Update the IP Address if Necessary:

If the displayed IP address is incorrect, use the following command to assign a new IP

```
> ipconfig 0 ip 10.255.255.6 255.255.255.252 10.255.255.5
```

This command will set the module's IP address to one that the server can connect to.

7. Verify the Connection:

In another terminal, run the command:

```
> detector_arpdisp
```

```
(det-env) lumentum@dcu-server-03:~$ detector_arpdisp
Module: 1      IP: 10.255.255.2      (enp23s0f0np0)  Status: Replied
Module: 2      IP: 10.255.255.6      (enp23s0f1np1)  Status: Replied
```

If both modules appear in **green**, the connection is successful, and you can proceed with the next commands as outlined in the instructions.

6.2. Change configuration to operate with one module

The system currently assumes that the user will consistently use the same detector model on the server. However, there are instances where the user needs to switch to a different model or operate with just a single module of the detector. In such cases, a bug has been identified that arises from the order in which the system initializes and its assumption of a persistent detector model. This causes a conflict between the memory of the previous configuration and the new one, leading to issues. This problem is particularly noticeable when the user attempts to acquire images or perform a new equalization using only one module after previously utilizing the complete detector.

To reproduce this issue, follow these steps:

1. Run the detector with only one module after consistently using it in full mode. In a console, execute:

```
> detector_console -m1
```

2. Initialize the camera server and load its configurations by running on the CLI:

```
> camera_start
```

This workflow represents the typical process for controlling the detector via the CLI. However, in this scenario, you'll notice that any acquisition attempts will fail. This is because the configuration still expects input from the other modules, which are not currently in use.

Fix

To resolve this issue, close the current console and repeat the steps:

1. Restart the detector console.
2. Run `camera_start` again in the new CLI session.

This ensures that the new configuration aligns with the current detector setup, allowing the system to function normally.

Additional Notes

The same bug and fix apply when switching from a single-module setup back to using the full detector. In that case:

1. Start the console for the full detector.
2. Start the camera server.
3. Restart the workflow by executing both commands.

This process ensures proper synchronization between the configuration and the detector state.

1. Operation Conditions

RAD detectors are made for indoor use only and should be operated only according to the following ambient conditions for each sensor type.

Ambient Conditions	Range	
	Silicon	CdTe
Temperature	> 20°C and < 35°C	> 20°C and < 30°C
Humidity	< 60% at 20°C, non-condensation	

Table 6: Ambient operation conditions.

Ensuring proper ambient conditions is essential to prevent sensor degradation caused by humidity and temperature fluctuations. Temperature, particularly critical for Medipix3RX chips, poses a significant concern as they are the most temperature-sensitive components of the detector. The maximum operating temperature for Medipix chips is detailed in the table below for each sensor type.

	Silicon	CdTe
Maximum Temperature	60°C	30°C

Table 7: Limit temperature for Medipix chips operation.

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The cooling system is designed to prevent the chips from exceeding these temperature limits to maintain optimal detector performance and functionality. However, if the limit is reached, the alarm system described below will promptly intervene. The temperature of the chips can be accessed through the software interface. Refer to the operating manual for detailed instructions on how to do so.

1.1. Alarm System

The detector is furnished with sensors designed to monitor ambient conditions, including temperature and humidity. Additionally, internal sensors are integrated to monitor the temperature of electronic boards and Medipix chips.

If the ambient temperature and humidity limits are exceeded, an alarm will trigger a disclaimer in the software interface, and the detector environment LED will turn red. Ensure that the appropriate ambient conditions are maintained to continue using the detector without risking equipment damage. Maximum performance is guaranteed only when operating within the specified conditions.

2. Storage Recommendations

When storing the detector, always make sure that the temperature and humidity inside the storage box do not exceed the conditions specified below. To control the humidity inside the box, always use a drying agent such as silica gel.

Storage Ambient Condition	Range
Temperature	> 20°C and < 35°C
Humidity	< 60% at 20°C, non-condensation

Table 8: Storage conditions.

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3. Handling Instructions

Make sure all personnel are trained in the handling of Electrostatic Sensitive Devices. Before handling the detector, ensure that all requirements for handling Electrostatic Sensitive Devices (ESD) are met. This includes ESD-certified laboratory equipment (e.g., tools, tables, chairs, ...) as well as ESD-labeled workwear.

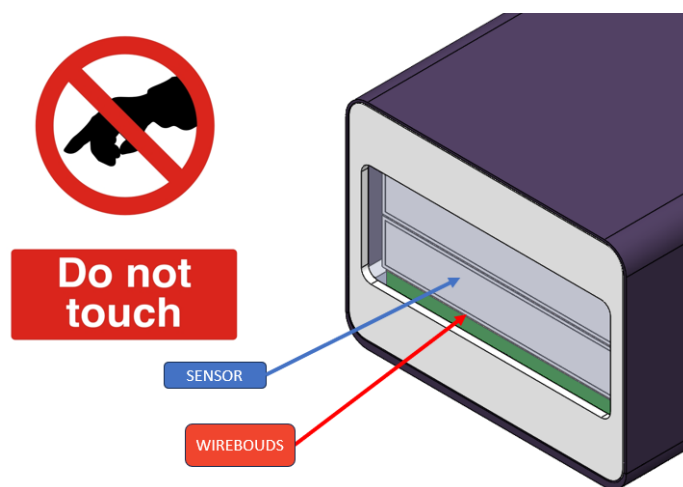


Figure 11: Handling precautions for the RAD Detector.

	Warning Do not touch the Mylar® covering foil, sensor or exposed wire bonds when the cover is removed.
	Warning The Mylar® covering foil may not protect the detector against radiation damage caused by direct incident synchrotron beam.

4. Maintenance

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The detector housing can be cleaned with soft tissue only.

	Warning
	Do not clean or touch the Mylar® foil covering located over the sensors' surface.

5. Warnings

	Warning
	Please, read these warnings carefully before operating the detector. • Do not touch the detector's optical head (sensors surface). • Place the protective cover on the optical head when the detector is not in use. • The detector is not built to withstand a direct beam at a synchrotron. Such exposure will damage the exposed sensors. • Operation outside the specified ambient conditions could damage the detector. • Power down the detector before connecting or disconnecting any cables. • Avoid any pressure or tension on the cables. • Opening the detector or the detector's power supply without explicit instructions from Lumentum. Ltd. will void the warranty. • The embedded operating system on the detector has customized software for controlling the RAD detector family systems. Do not make any changes to the operating system without explicit instructions from Lumentum ltda.

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